r/aiwars

# How to make AI more sustainable in the long run?

Posted by u/Icy-Humor2907 • • 0 points • 19 comments

A huge point of contention between both sides is the environmental impacts that generative AI has, specifically regarding the large amounts of water needed to cool the data centers that hold the foundations for LLMs or image generators. Excluding data centers that run non-AI services, AI does still require a significant amount of water. For example, producing a singular AI microchip requires 2.1-2.6 gallons (8-10 liters) of water. This is also not mentioning the amount needed to cool or even power data centers for AI.

For another reference, only .5% of the world’s water supply is fresh water that is accessible to humans. The actual amount of fresh water is roughly 3%, however most of that is locked in ice caps; meaning there is simply not a lot of water to spare, especially when 10% of the planet is living in water scarcity.

The question is, how can that be combatted? If the current method of using water isn’t a long-term sustainable option, what other options are available? Air cooling? Solar power? Wind power? I don’t believe AI alone is killing the planet (data centers running things like Gmail and Google Drive use 550,000 gallons of water per day); however, any little bit counts towards the continued livability of Earth.

All opinions welcome.

## Comments

> **u/Feroc** • 1 points • 2025-08-20 21:31 UTC
>
> https://preview.redd.it/xvn6y863s8kf1.png?width=1000&format=png&auto=webp&s=c0d86e0ebb108b85a5f2401d18f7d863f4d43f75
>
> There are no numbers for 2025, but with the 2015 numbers from the USGS and the 2024 numbers from Lawrence Berkeley National Lab, we are talking about a literal drop on a hot stone. The USGS will release new numbers sometime this year. Surely, data centers will need more than they did in 2015, but if the goal is to save water, then we will surely find better areas to focus on.

> **u/NegativeKitchen4098** • 2 points • 2025-08-20 19:23 UTC

It's a waste of time. If you want to conserve water for the environment, cut back a little on the number of almonds you eat, and burgers.

1 almond = 12l of water (https://www.sciencedirect.com/science/article/pii/S1470160X17308592)

1 1/4lb burger = 1750l of water https://water.usgs.gov/edu/activity-watercontent.php

There are so many other things we can do to save water, that it's ridiculously ineffective to do so by targeting AI / computer use.

> **u/FadingHeaven** • 1 points • 2025-08-21 02:50 UTC
>
> It's not a waste of time. Reducing water and energy usage saves money so it's not like the AI companies gain nothing from this. Also as they become more prevalent their impact is gonna increase.
>
> I doubt you'd be happy if your city had rolling blackouts cause of the strain on the power grid or worse air quality cause of the natural gas they burn.
>
> Almond manufacturers can't do a thing about how much water an almond tree needs to grow. A single person eating less almonds doesn't stop that water from being used. Even if demand is lower they're not just gonna let their trees die as a result.
>
> AI companies can single-handedly reduce their impact without having to hope consumers change their habits to do so. Google had carbon neutral data centres for years. Closed loop systems should be a standard. Making chips more energy efficient should be a goal too.

> **u/Icy-Humor2907** • 2 points • 2025-08-20 19:24 UTC
>
> Thankfully for me, I already dislike almonds. I can also live without burgers.
>
> > **u/nomic42** • 1 points • 2025-08-20 20:36 UTC
> >
> > But coffee? I'm not sure I'm ready for this. Fortunately, data centers aren't the biggest problem here.

**u/ShagaONhan** • 1 points • 2025-08-20 19:19 UTC

322 billion gallons per day in US 0.8 billions for all the data centers combine. That's 0.2% of the total, why nobody panic for the other 99.8% ? That can be improved by recirculating the water or not placing them in stressed environment. The water is not disintegrated, In a non stressed environment you are going to get it back in the cycle.

Even if they are evil corporations they want to save money, they are not going to waste resources for the sake of it. Add some fines on top and make them pay for the resources and we end up with very efficient systems.

**u/Kathilliana** • 4 points • 2025-08-20 19:07 UTC

Why are we discounting crypto, streaming, gaming, Wall Street and others in our calculation? They are the biggest piece of the pie, still.

A spike in the math happens in the prompt. While the user is reading the output, connection to data center goes down to zero. When you fall asleep on the couch with Netflix on? Still talking to the data center, wasting water. The moment I stop interacting with the LLM, I’m no longer wasting data center.

> **u/Icy-Humor2907** • 0 points • 2025-08-20 19:10 UTC

I was specifically asking about sustainability regarding AI, but I made a point of mentioning how hyperdata centers use significantly more water than AI does. I'm trying to at least somewhat talk about both sides of the argument.

**u/Kathilliana** • 6 points • 2025-08-20 19:19 UTC

I just hear that a lot. "We can't count movies, tv, songs, crypto, Wall Street, games" or any stuff we use right now. It's THIS one that's going to destroy us.

A person engaging in a FPS game or mining crypto is using way more power at the data center than someone having drawn out conversations with an LLM. We spend way more time reading than actively having the LLM calculate.

**u/No-Opportunity5353** • 3 points • 2025-08-20 19:48 UTC

Fun fact: generating images actually uses *less* power than generating text.

**u/Hegel93** • 2 points • 2025-08-20 19:03 UTC

While server farms do require water, they don't necessarily need fresh water. You can use waste water and even sea water.

**u/Icy-Humor2907** • 1 points • 2025-08-20 19:06 UTC

Would that require purification due to the potential negative effect the salt or waste concentration in the water would have? I'm genuinely asking

**u/ShagaONhan** • 1 points • 2025-08-20 19:26 UTC

Works for nuclear plants. You can have indirect heat exchange by using a secondary loop. And since it's not has hot that make it even easier.

**u/Hegel93** • 1 points • 2025-08-20 19:24 UTC

I imagine you have to do some filtering just so large matter doesn't go through and use material and seals that won't corrode due to salt but it's been done with fairly little controversy. there would still be some environmental concerns with like... taking in tepid water and releasing warm water etc. but it's not one you have to run these things off tap water.

**u/cultofmilkandhoney** • 2 points • 2025-08-20 18:59 UTC

The world is moving towards using renewable resources for energy. China knows this and is capitalizing on it. The entire environment argument is DUMB.

**u/cultofmilkandhoney** • 2 points • 2025-08-20 19:00 UTC

Anti-AI people have 0 idea what they are talking about. Do not engage.

**u/MysteriousPepper8908** • 2 points • 2025-08-20 18:54 UTC

Model optimization which we're already seeing will allow the increased usage of AI to result in minimal increases in water usage or omissions but none of these figures mean anything because they're not presented in context. 8 liters of water is meaningless unless you understand how much water exists to be expended and the relative water use of other industries and the reality is AI is a drop in the proverbial bucket and will remain so for the foreseeable future. AI can also improve the efficiency of data centers itself as it already has by finding a more efficient algorithm for matrix multiplication, something that data centers are doing trillions of times a day.

**u/Icy-Humor2907** • 1 points • 2025-08-20 19:01 UTC

Hence why I mentioned the comparative amount of water other data centers such as Gmail and Google Drive use. I also don't believe 8 liters of water is meaningless, considering the average person needs roughly 1.7 liters of water a day (7 cups). The 8 liters used to produce a single microchip could give almost 5 people water for a day. So while yes, 8 liters is a drop in the ocean compared to available water, it is still a sizable amount of water (hence why I mentioned the long run, not the short term)

Also, model optimization is a good point. Optimization means less space required, and therefore less water required. Though do you believe optimization will eventually lead to further mass production in order to keep up with the increasing demand for accessible, fast AI?

**u/MysteriousPepper8908** • 1 points • 2025-08-20 19:10 UTC

Another example of how figures out of context are meaningless. People needing drinking water is not a serious burden on the water supply, lack of clean water is more of an issue of pollution or a local issue in very arid regions, AI isn't threatening access to drinking water. You might as well compare it to to the amount of water required to satiate a butterfly and look, AI is a billion times more than that so it must be really bad! Without recognizing that orders of magnitude more water are going to things like social media and pistachio farming but we're going to carve out some special outrage over AI because we prefer pistachios to AI. AI is already being used by over a billion people on a weekly basis and while amount of usage per person will likely increase, there are realistic caps on how much utilization it can reasonably receive relative to where it is now.